MODULE 2

# Hydrogen Production Technologies

Nine routes – principle, reactions, equipment, maturity

**How to read this module**
Each technology follows the same card: **principle → reactions → flow → equipment → conditions → energy → purity → pros/cons → applications**. The coloured bar shows maturity: **green = commercial**, **orange = demonstration / early commercial**, **red = laboratory / pilot**. Where a reliable public figure was not found, the parameter is described qualitatively rather than invented.

## 2.1 Steam Methane Reforming (SMR)

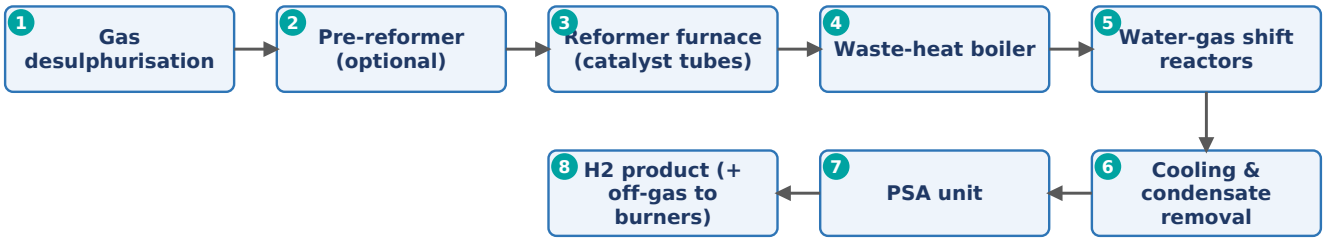
**Technology maturity:** Commercial

**Working principle.** Methane reacts with high-temperature steam over a nickel-based catalyst in tubes inside a fired furnace. The reaction is endothermic, so heat is supplied by burning fuel (natural gas plus PSA off-gas). A water-gas shift step then converts CO into more $H_2$, and pressure swing adsorption purifies the product [R2].

$$CH_4 + H_2O + \text{heat} \rightarrow CO + 3H_2$$
Reforming (endothermic, ΔH ≈ +206 kJ/mol)

$$CO + H_2O \rightarrow CO_2 + H_2 + \text{small heat}$$
Water-gas shift (exothermic, ΔH ≈ –41 kJ/mol)

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Natural gas (or LPG, naphtha, biogas), process steam (boiler feed water), combustion air |
| **Main equipment** | Desulphuriser (ZnO beds), reformer furnace with catalyst-filled tubes and burners, waste-heat/steam boiler, HT/LT shift reactors, process gas coolers, condensate separators, PSA adsorber beds, product compressor; for blue $H_2$: $CO_2$ capture unit |
| **Operating conditions** | Reformer: **700 – 1,000 °C** at **3 – 25 bar** with catalyst [R2]. Shift: lower temperature, catalytic. |
| **Energy requirement** | Thermal energy (furnace firing) dominates; natural gas is both feedstock and fuel. |
| **Hydrogen purity** | PSA delivers essentially pure hydrogen [R2] (the $CO_2$ and impurities are removed). |
| **Advantages** | Mature, large single-train capacity, lowest cost where gas is cheap, huge operating experience |
| **Disadvantages** | $CO_2$ emissions 10–14 kg $CO_2$-eq/kg $H_2$ unabated [R5]; flue-gas $CO_2$ is dilute and costly to capture |
| **Industrial applications** | Refineries, ammonia and methanol plants, merchant hydrogen |

## 2.2 Autothermal Reforming (ATR)

**Technology maturity:** Commercial

**Working principle.** ATR combines partial oxidation and steam reforming inside one refractory-lined reactor. Oxygen and steam are fed with the gas; the exothermic oxidation supplies the heat for the endothermic reforming, so no external furnace is needed. Because there is no fired furnace, almost all $CO_2$ ends up in the high-pressure process stream – which makes carbon capture easier. ATR is therefore the preferred route for many large **blue hydrogen** projects.

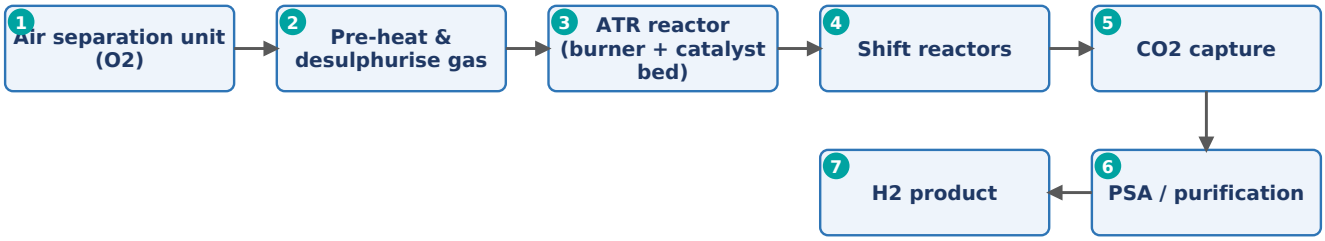
$$CH_4 + ½O_2 \rightarrow CO + 2H_2 \text{ (+ heat)}$$

Partial oxidation zone

$$CH_4 + H_2O \rightarrow CO + 3H_2 \text{ (uses that heat)}$$

Catalytic reforming zone, then water-gas shift

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Natural gas, oxygen (from an air separation unit), steam |
| **Main equipment** | Air separation unit, ATR reactor with burner and catalyst bed, waste-heat boiler, shift reactors, amine (or similar) $CO_2$ capture, PSA |
| **Operating conditions** | High-temperature, oxygen-blown; specific temperatures and pressures are licensor-specific (not given here to avoid unsourced values). |
| **Energy requirement** | Self-heated reactor; electricity for the air separation unit is a significant load. |
| **Hydrogen purity** | High after shift + PSA / purification |
| **Advantages** | No furnace; concentrated $CO_2$ stream; large single-train size; well suited to CCS |
| **Disadvantages** | Needs an oxygen plant (capital + power); still fossil-based; CCS needs storage site |
| **Industrial applications** | Large ammonia/methanol plants; blue hydrogen hubs |

## 2.3 Partial Oxidation (POX)

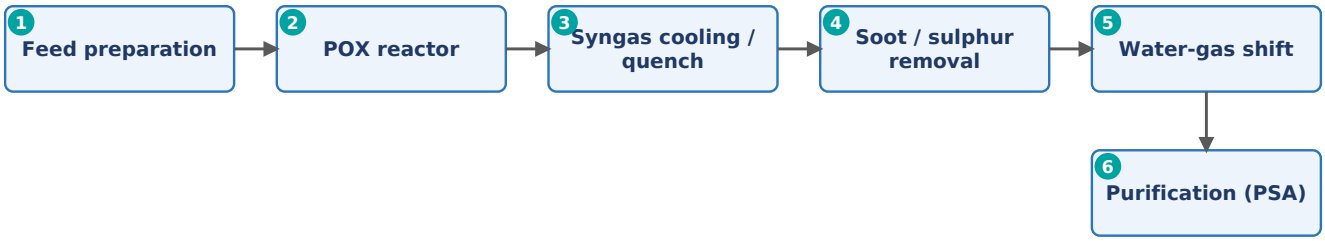
**Technology maturity:** Commercial

**Working principle.** Methane or heavier hydrocarbons react with a limited amount of oxygen. The reaction is exothermic, typically faster than steam reforming and needs a smaller reactor, but gives less hydrogen per unit of fuel [R2]. Shift and purification follow as in SMR.

$$CH_4 + ½O_2 \rightarrow CO + 2H_2 + \text{heat}$$

Exothermic; air as oxidant adds N2 to the product [R2]

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Natural gas, heavy oil residues, petroleum coke; oxygen or air |
| **Main equipment** | Oxygen plant, refractory POX gasifier, quench or syngas cooler, soot scrubber, acid-gas removal, shift reactors, PSA |
| **Operating conditions** | Non-catalytic, high-temperature operation; conditions depend on feedstock and licensor. |
| **Energy requirement** | Self-heated; oxygen production is the main parasitic energy load. |
| **Hydrogen purity** | Syngas contains CO, $CO_2$, some $N_2$ (if air-blown) – purified by shift + PSA |
| **Advantages** | Handles heavy and dirty feedstocks; compact, fast reaction |
| **Disadvantages** | Lower $H_2$ yield than SMR; oxygen plant cost; $CO_2$ emissions |
| **Industrial applications** | Refinery residue upgrading, syngas for chemicals |

# 2.4 Water Electrolysis (overview)

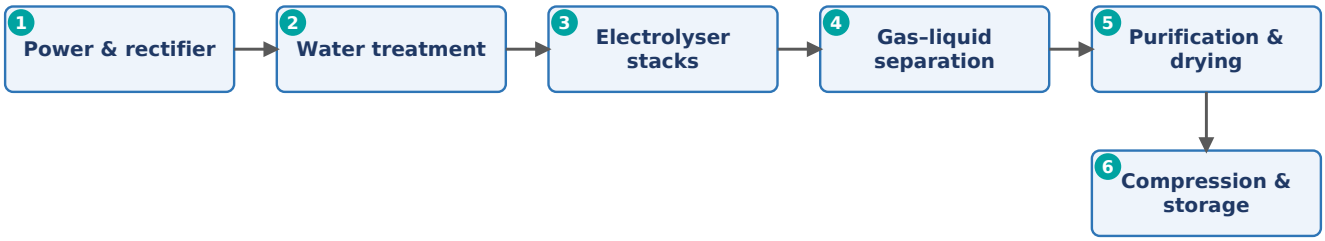
**Technology maturity:** Commercial

**Working principle.** Direct current splits water into hydrogen at the cathode and oxygen at the anode. With renewable or nuclear electricity the process emits no $CO_2$ at the production stage. Four cell types exist – alkaline (AEL), proton exchange membrane (PEM), solid oxide (SOEC) and anion exchange membrane (AEM) – detailed in Module 4.

$$2H_2O + \text{electricity} \rightarrow 2H_2 + O_2$$

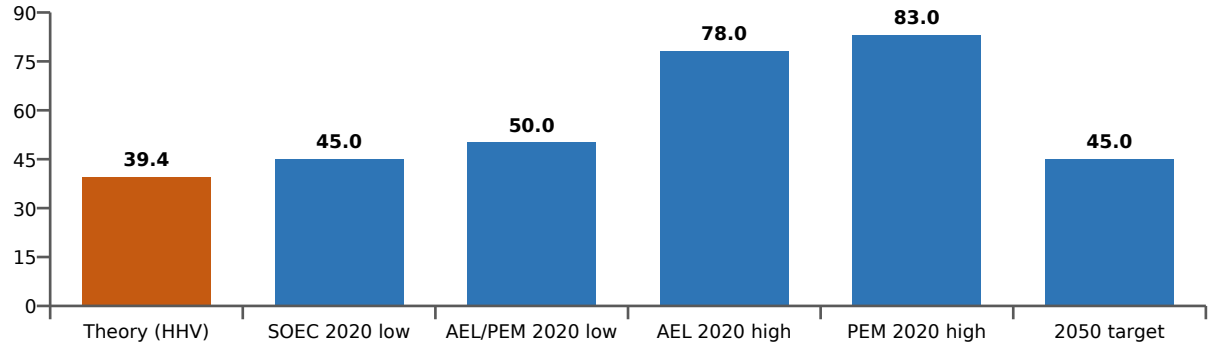
ΔH = 285.8 kJ/mol H2 (HHV) · reversible voltage 1.23 V · thermoneutral voltage 1.48 V (25 °C)

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Demineralised water (9 kg per kg $H_2$ stoichiometric; 18 – 24 kg incl. demineralisation losses [R1]); DC electricity |
| **Main equipment** | Transformer–rectifier, stacks, separators, circulation pumps, heat exchangers, de-oxo, dryer, compressor |
| **Operating conditions** | AEL 70–90 °C / 1–30 bar; PEM 50–80 °C / <70 bar; AEM 40–60 °C / <35 bar; SOEC 700–850 °C / ~1 bar [R1] |
| **Energy requirement** | System 2020: AEL 50–78, PEM 50–83, AEM 57–69, SOEC 45–55 kWh/kg $H_2$; 2050 target <45 (SOEC <40) [R1] |
| **Hydrogen purity** | Very high after de-oxo and drying – suitable for fuel cells |
| **Advantages** | Zero direct emissions with clean power; modular; flexible operation (esp. PEM); $O_2$ by-product |
| **Disadvantages** | Electricity cost dominates; stack capital cost; critical materials (Ir, Pt for PEM) |
| **Industrial applications** | Green ammonia, refineries, steel DRI, refuelling stations, energy storage |

**Electricity per kg H2 (kWh/kg) – theory vs. system (IRENA, 2020)**




*Theoretical minimum = HHV 39.4 kWh/kg. Real systems need more because of overpotentials, ohmic losses and balance-of-plant loads [R1].*

## 2.5 Biomass Gasification

**Technology maturity:** Early commercial / demonstration

**Working principle.** Biomass is heated with a controlled amount of oxygen and/or steam (without full combustion) to make syngas ($H_2$, CO, $CO_2$, $CH_4$). Tars and particulates are removed, CO is shifted to $H_2$, and the gas is purified.

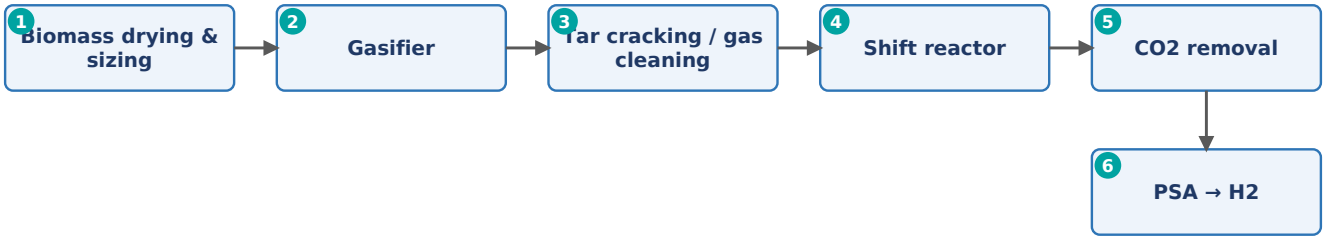
$$\text{Biomass} + O_2 / H_2O \rightarrow CO + H_2 + CO_2 + CH_4 + \text{tar}$$

Gasification

$$CO + H_2O \rightarrow CO_2 + H_2$$

Water-gas shift

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Agricultural residues, wood chips, energy crops, municipal solid waste (bio-fraction) |
| **Main equipment** | Feed handling, dryer, gasifier (fixed bed / fluidised bed / entrained flow), cyclones, tar reformer, scrubbers, shift, PSA |
| **Operating conditions** | Gasifier temperature and pressure depend on design; tar control is the key operating issue. |
| **Energy requirement** | Thermal energy from partial oxidation of the biomass itself |
| **Hydrogen purity** | High after PSA |
| **Advantages** | Uses waste/residue; can be carbon-neutral or negative with CCS |
| **Disadvantages** | Variable feed quality; tars foul equipment; logistics of collecting biomass; smaller scale |
| **Industrial applications** | Distributed hydrogen near agri-residue sources; waste-to-hydrogen |

## 2.6 Coal Gasification

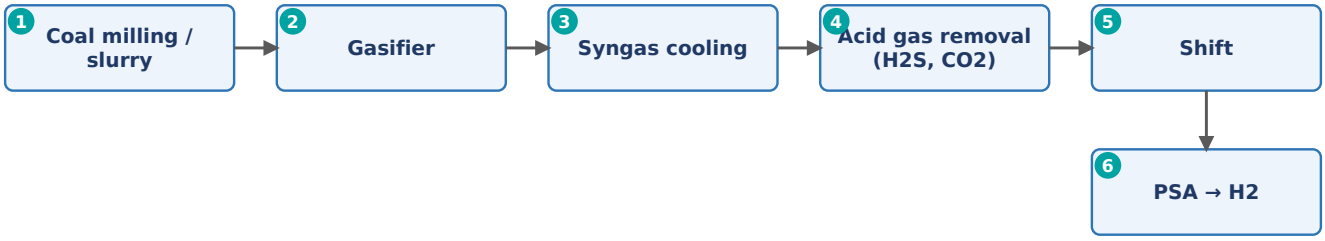
**Technology maturity:** Commercial

**Working principle.** Coal is gasified with oxygen and steam in an entrained-flow or fluidised-bed gasifier to syngas, which is cleaned, shifted and purified. Widely used in China for ammonia and methanol.

$$C + H_2O \rightarrow CO + H_2$$
Steam gasification (endothermic)

$$C + \tfrac{1}{2}O_2 \rightarrow CO$$
Partial combustion supplies heat

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Coal or lignite, oxygen, steam |
| **Main equipment** | Air separation unit, gasifier, syngas coolers, acid-gas removal, sulphur recovery, shift reactors, PSA |
| **Operating conditions** | High-temperature, oxygen-blown, pressurised gasifiers (design-specific) |
| **Energy requirement** | Large; plus oxygen production |
| **Hydrogen purity** | High after purification |
| **Advantages** | Uses abundant domestic coal; mature large plants |
| **Disadvantages** | Highest $CO_2$ intensity of all routes (up to ≈27 kg $CO_2$-eq/kg incl. upstream [R5]); ash and water handling |
| **Industrial applications** | Ammonia, methanol, coal-to-chemicals |

## 2.7 Methane Pyrolysis (turquoise hydrogen)

**Technology maturity:** Early commercial / demonstration

**Working principle.** Methane is cracked at high temperature without oxygen into hydrogen and **solid carbon**. No $CO_2$ is formed in the reaction; the carbon (carbon black, graphite) can be sold. Heat comes from plasma, molten metal/salt baths or catalysts.

$$CH_4 + \text{heat} \rightarrow C(s) + 2H_2$$

ΔH ≈ +75 kJ/mol CH4 – much less energy per kg H2 than electrolysis

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Natural gas or biomethane; electricity or heat |
| **Main equipment** | Plasma torch reactor or molten-metal bubble column, carbon separation (cyclones, filters), gas clean-up / PSA |
| **Operating conditions** | Very high temperature; design-specific |
| **Energy requirement** | Lower electricity per kg $H_2$ than electrolysis (theoretical) |
| **Hydrogen purity** | High after clean-up |
| **Advantages** | No process $CO_2$; saleable carbon by-product |
| **Disadvantages** | About 40 % of the methane's heating value stays in the solid carbon (802 vs 2×242 kJ/mol LHV); carbon market size; reactor fouling; upstream methane leaks still count |
| **Industrial applications** | Carbon black + hydrogen co-production |

## 2.8 Photoelectrochemical (PEC) Water Splitting

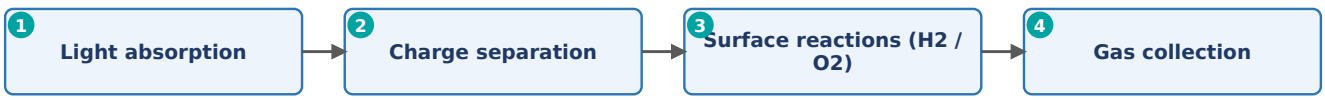
**Technology maturity:** Laboratory

**Working principle.** A semiconductor electrode immersed in water absorbs sunlight and directly drives water splitting – combining a solar cell and an electrolyser in one device.

$$2H_2O + \text{sunlight} \rightarrow 2H_2 + O_2$$

Photo-generated electron–hole pairs drive the half-reactions

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Water, sunlight, semiconductor photoelectrodes |
| **Main equipment** | Photo-electrodes (metal oxides, III-V semiconductors), membranes, transparent reactor panels |
| **Operating conditions** | Ambient temperature, sunlight |
| **Energy requirement** | Solar energy only |
| **Hydrogen purity** | Requires drying/purification |
| **Advantages** | Direct solar-to-hydrogen; potentially simple |
| **Disadvantages** | Low efficiency and durability (corrosion) today; not yet scaled |

| Item | Details |
|---|---|
| **Industrial applications** | Research |

## 2.9 Biological Hydrogen Production

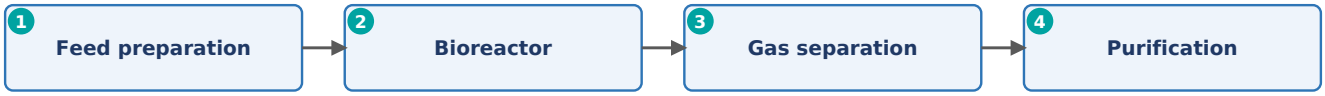
**Technology maturity:** Laboratory / pilot

**Working principle.** Micro-organisms produce hydrogen: algae and cyanobacteria by **photobiological** water splitting, and bacteria by **dark fermentation** of sugars/organic waste or photo-fermentation.

$$C_6H_{12}O_6 + 2H_2O \rightarrow 2CH_3COOH + 2CO_2 + 4H_2$$

Dark fermentation (acetate pathway)

**Process flow**




| Item | Details |
|---|---|
| **Raw materials** | Organic waste, wastewater, sugars; sunlight for photobiological routes |
| **Main equipment** | Bioreactors / photobioreactors, gas separation membranes, PSA |
| **Operating conditions** | Near-ambient temperature |
| **Energy requirement** | Low external energy |
| **Hydrogen purity** | Gas contains $CO_2$; must be purified |
| **Advantages** | Uses wastes at low temperature |
| **Disadvantages** | Low production rates and yields; large reactor volumes |
| **Industrial applications** | Research; possible niche in waste treatment |

## 2.10 Summary comparison

| Route | Feedstock | Direct $CO_2$ | Maturity | Main limitation |
|---|---|---|---|---|
| **SMR** | Natural gas + steam | Yes (dilute) | Commercial | Emissions |
| **ATR** | Gas + O2 + steam | Yes (concentrated) | Commercial | Needs CCS + O2 plant |
| **POX** | Gas / heavy oil + O2 | Yes | Commercial | Lower yield |
| **Electrolysis AEL/PEM** | Water + power | No | Commercial | Power cost |
| **Electrolysis SOEC/AEM** | Steam/water + power | No | Lab-scale (IRENA 2020) → scaling | Durability, scale |
| **Biomass gasification** | Biomass | Biogenic | Demonstration | Tars, logistics |
| **Coal gasification** | Coal | Yes (highest) | Commercial | Emissions |
| **Methane pyrolysis** | Gas | No (solid C) | Early commercial | Carbon market, scale |
| **PEC** | Water + sun | No | Laboratory | Efficiency |
| **Biological** | Waste / sun | Biogenic | Laboratory / pilot | Rate |

# Quiz & revision – Module 2

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

**Q1.** SMR reformer conditions per US DOE:
**a)** 200–300 °C, 100 bar **b)** 700–1,000 °C, 3–25 bar **c)** 1,500 °C, 1 bar **d)** 80 °C, 30 bar

**Q2.** Why is ATR preferred for blue hydrogen?
**a)** Uses no oxygen **b)** CO2 is concentrated in the process stream **c)** Has no catalyst **d)** Produces no CO

**Q3.** By-product of methane pyrolysis:
**a)** CO2 **b)** Solid carbon **c)** Ammonia **d)** Oxygen

**Q4.** Stoichiometric water per kg H2:
**a)** 1 kg **b)** 4.5 kg **c)** 9 kg **d)** 24 kg

**Q5.** Which route is still laboratory-stage?
**a)** SMR **b)** Coal gasification **c)** PEC water splitting **d)** PEM electrolysis

**Answer key**
**Q1 – b**: DOE [R2]; **Q2 – b**: no fired furnace, capture is easier; **Q3 – b**: carbon black/graphite; **Q4 – c**: 18 g water → 2 g H2; **Q5 – c**: photo-electrochemical